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Source: *Child Development*, Aug., 1994, Vol. 65, No. 4 (Aug., 1994), pp. 1028-1041

Published by: Wiley on behalf of the Society for Research in Child Development

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Responses of Free-ranging Rhesus Monkeys to a Natural Form of Social Separation. I. Parallels with Mother-Infant Separation in Captivity

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BERMAN, C. M.; RASMUSSEN, K. L. R.; and SUOMI, STEPHEN J. *Responses of Free-ranging Rhesus Monkeys to a Natural Form of Social Separation. I. Parallels with Mother-Infant Separation in Captivity.* CHILD DEVELOPMENT, 1994, **65**, 1028–1041. Observations of 23 free-ranging rhesus monkey infants on Cayo Santiago, Puerto Rico, indicated that mothers' first postpartum estrous periods were marked by large increases in the amount of time infants were separated from their mothers, by disturbances in mother-infant relationships, and by increases in infant distress behavior. When their mothers resumed mating, most infants showed signs of agitation; a few briefly showed indications of depression. Male infants responded to their mothers' resumption of mating by playing more, whereas females engaged in less play and more allogrooming. The results suggest (*a*) that basic parallels exist between the behavioral responses of rhesus infants to their mothers' resumption of mating in the field and to forcible separation from their mothers in captivity and (*b*) that early separation experiences may play a role in the normal development or manifestation of sex differences in behavior.

Nearly 3 decades ago, comparative psychologists and ethologists began to document the agitation and/or depressive responses of infants from a wide range of nonhuman primate species to forcible separation from their mothers (see reviews by Coe, Weiner, Rosenberg, & Levine, 1985; Hinde & McGinnis, 1977; Kraemer, Ebert, Lake, & McKinney, 1983; Mineka & Suomi, 1978). Such studies have figured heavily in theories of parent-offspring attachment (e.g., Ainsworth, 1979; Bowlby, 1969; Rajecki, Lamb, & Obmascher, 1978) and the etiology of affective disorders in humans (e.g., Kaufman & Rosenblum, 1967; McKinney & Bunney, 1969; McKnew & Cytryn, 1973; Suomi, 1991). However, some researchers have questioned the generality of these studies, because factors such as housing conditions, rearing conditions, and group compositions were often atypical for the species under investigation (e.g., Ainsworth, 1976). Hence, the extent to which these laboratory findings model normal developmental phenomena is still controversial.

Our casual observations of free-ranging rhesus monkeys (*Macaca mulatta*) suggested that infants commonly experience repeated short-term separations from their mothers when their mothers resume mating and that many infants respond to these separations with marked distress and disturbance. Hence, the natural events accompanying the mother's resumption of mating may parallel those seen in captive studies of mother-infant separation. Indeed, infants' ages when mothers resume mating and mothers' periods of intensive mating tend to be similar to infants' ages and periods of separation in many captive studies. On the other hand, there are clear differences between

The authors thank John Berard and Matt Kessler for permission to work on Cayo Santiago and for logistical support; Barbara Welker, Stacey McCulle, and Ellen Kapsalis for assistance with the behavioral observations; and Michael Hulett and Helen Ball for computer programming. Robert Hinde made useful comments on an earlier version of the manuscript. Funding for this project was provided by a grant from the H. F. Guggenheim Foundation and by intramural support from the National Institute of Child Health and Human Development. Please send reprint requests to Dr. C. M. Berman, Department of Anthropology, State University of New York at Buffalo, Buffalo, NY 14261.

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the two situations. First, mothers in captive studies are generally forcibly removed from their infants (or vice versa), whereas mothers in the field "voluntarily" make themselves unavailable during estrus. Second, separation in the field is rarely as complete as it is in the laboratory. Last, infants in naturally organized social groups may be able to interact with and gain comfort from a larger and more varied set of kin and peers (e.g., Mineka & Suomi, 1978) than infants in most captive settings. Since responses to separation have been found to vary with the physical and social context (Hinde & McGinnis, 1977; Mineka & Suomi, 1978), it is important to ask whether infants in the field display qualitatively or quantitatively different patterns of response to separation relative to those in captivity.

To date, only Singh (1975) has studied temporary maternal separation in the field. Five infant rhesus macaques who were forcibly separated from their social groups showed protest and mildly depressive responses, as indicated by decreases in locomotion, vocalization, exploration, and feeding. Studies of permanently orphaned infants in the field have described depressed behavior usually followed by death, particularly for orphans who were not adopted by group companions (Berman, 1982; Hasegawa & Hiraiwa, 1980; Rhine, Norton, Roertgen, & Klein, 1980). Worlein, Eaton, Johnson, and Glick (1988) found increased frequencies of maternal rejections and infant distress among corral-housed Japanese macaques during the mating season. Gomendio (1991) found similar changes among 12 captive rhesus mother-infant pairs during the month that included the mother's first estrus period. However, neither study examined the degree to which these changes were linked specifically to the mating activity of individual mothers and/or to the separation of infants from mothers.

Here we document the extent to which free-ranging infant rhesus monkeys on Cayo Santiago, Puerto Rico, experience increased separation from their mothers when their mothers resume mating, compare mother-infant relationships during the mother's first intensive mating period with those seen immediately before it, and examine changes in infant distress-related behavior, social play, and grooming that accompany the mother's resumption of mating. Since responses to separation have been found to vary with the quality of the mother-infant relationship and with a number of attributes of the mother or infant, the possible importance of a number of factors (infant's sex, infant's age, mother's dominance status, and numbers of sisters and other close kin) thought to play important roles in free-ranging rhesus mother-infant relationships (Berman, 1984, 1992) and/or in reactions to separation in captivity (Hinde & McGinnis, 1977; Mineka & Suomi, 1978) is also examined. More detailed descriptions of individual differences in responses to the mother's resumption of mating (Berman, Rasmussen, McCulle, & Suomi, in press) will be presented elsewhere.

Methods

Study Site and Population

Cayo Santiago is a 37-acre island off the east coast of Puerto Rico that supports a free-ranging, provisioned, predator-free, seasonally breeding population of rhesus monkeys. During the course of the study, there were six naturally formed, species-typical social groups and a small number of peripheral males, comprising a total population of about 1,200 individuals. The population has been censused daily since 1956 and hence was well-habituated and made up of known individuals with known histories and maternal kinship relationships. Since 1972, management practices have been designed to minimize human disturbance and have been virtually constant. Detailed descriptions of the history, terrain, population, and management of Cayo Santiago are given in Rawlins and Kessler (1983) and Sade, Chepko-Sade, Schneider, Roberts, and Richtsmeier (1985).

Study Groups and Subjects

Each study group (F, Q, and T) was organized in a manner typical of multimale, multifemale macaque groups (Lee, 1983). As in their natural habitats in India, female rhesus on Cayo Santiago remain in their natal groups for life and associate closely with their mothers and maternal kin. Males, on the other hand, generally leave their natal groups around puberty and join other groups. In this way, large maternal kinship lineages, which are also subunits of social organization, develop within groups on Cayo Santiago. During the study, groups F, Q, and T had four, three, and two such lineages, respectively, and total memberships of about 284, 182, and 128, respectively. The dominance structure of each group was closely linked with the kinship structure: mature daughters ranked immediately below their mothers, and lineages as a whole ranked in the same order as their matriarchs.

Births generally take place between De-

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cember and May on Cayo Santiago, and mating typically resumes in July and continues until November. A sample of 23 infants (all born during the 1990 birth season) was observed from the beginning of June to mid-November 1990: Eight each were from groups F and Q, and seven were from group T. Subjects were selected so as to match and balance the samples in each group as far as possible for the infant's sex, the mother's age (5–6, 7–9, 10–12, >12 years), and the mother's dominance rank. Dominance relationships were assessed by recording cowers and fear grins between pairs of adult females on an ad libitum basis (de Waal & Lutrell, 1985; Sade, 1967). Mothers were classified as high-ranking if they were members of the top-ranking lineage in their group and low-ranking if they were members of the other lineages. All the mothers were multiparous.

Data Collection

Intensive mating in group-living rhesus mothers is a reliable indicator of estrus (Gordon, 1981; Herndon, Turner, Ruiz de Elvira, & Collins, 1987). In order to track reproductive states, each mother was monitored several times each day for evidence of mating. A female was considered to have mated if she was observed in a series mount with an adult or subadult male or if she displayed a vaginal plug of coagulated semen.

The infant observation schedule was designed to maximize probabilities of observing infants on days on which they were most likely to experience traumatic rejections or separations from mothers. Infants were observed for three 15-min sessions per week until their mothers began to mate. At that time, the number of observation sessions was increased to twice daily until the mother was no longer observed with signs of mating for 4 consecutive days. To the extent possible, observation sessions for each infant were distributed equally among four daily time periods both before and after the mother began mating: 0700–0930, 0930–1200, 1300–1530, and 1530–1800 hours AST. Except for this constraint, the sequence of infants observed was determined by a random order arranged at the beginning of each week.

Focal-animal sampling methods (Altmann, 1974) were used to record the frequencies, durations, and sequences of all interactions of the infant with its mother and with other group members. Interactions were recorded chronologically on hand-held computers (Micropalm 4000/5000). At 3-min intervals during each focal-animal sampling session, point time samples were collected noting the infant's activity, the identity of animals in contact with and in proximity to the infant, and the infant's apparent behavioral state. There was a high level of interobserver reliability between the senior author and each of the three other data collectors (kappa coefficient $\geq .90$; Cohen, 1960) for the measures analyzed here. Table 1 lists the behaviors and their definitions.

Data Analysis

In order to distinguish changes in behavior that were linked specifically to the events surrounding the mother's resumption of mating from general responses to the mating season, we compared measures during each mother's first estrous period with those for a 5-week baseline period immediately preceding it. The baseline data for nearly all infants in the sample were recorded after the onset of mating within the infant's social group, and in several cases, during the peak of mating activity. Estrous scores for each infant represented observations collected during the first period in which the mother was observed with signs of mating at least every other day. This period lasted 4–9 days, $M \pm SD = 7.4 \pm 1.5$, for all but one mother. This mother mated intensively for 15 days, but only the first 9 days were included in the analyses.

The statistical analyses were performed using SPSS 4.1 (SPSS, Inc., 1990). We used repeated-measures multivariate analysis of variance (MANOVA) methods to assess the significance of behavioral changes between the baseline and estrous periods. Infant sex, age, and group membership were controlled by entering them as factors (sex and group) or covariates (age). Separate analyses were performed for measures of mother-infant interaction, infant distress, and infant social interaction with other individuals (see Table 1). Where necessary, arcsine and/or square-root transformations were applied to behavioral measures to meet the assumptions for parametric analysis. The relative contribution of each behavioral measure to the changes found was examined subsequently in a series of t tests for related samples.

Possible sources of interindividual variation in behavior during the baseline and estrous periods were assessed using analysis of covariance (ANCOVA) methods. Because the sample size, $N = 23$ infants, did not permit the analysis of all independent variables in a single design, we initially analyzed the

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TABLE 1

BEHAVIORS ANALYZED AND DEFINITIONS

Measures of Mother-Infant Interaction
Time on nipple: Percentage of point time samples on the mother's nipple
Time in contact: Percentage of point time samples in any kind of bodily contact with the mother
Time carried: Minutes per hour of observation time being carried either dorsally or ventrally by the mother
Time <60 cm: Percentage of point time samples in which the mother is within 60 cm (i.e., touching distance) of the infant
Time >5 m: Percentage of point time samples in which the mother is in sight but more than 5 m from the infant
Time out of sight: Percentage of point time samples in which the mother is apparently out of sight of the infant
Punishment rate: Frequency per hour with which the mother hits, bites, slaps, or pushes infant away
Contact initiation: Proportion of nipple contacts initiated by the mother divided by the total nipple contacts initiated by mother and infant
Rejection rate: Proportion of attempts to get on the nipple that are prevented by the mother divided by the total number of attempts initiated by mother or infant
Nipple attempts: Frequency per hour of successful and rejected attempts by infants to gain access to the mother's nipple
Nipple contacts: Frequency per hour of successful attempts to gain access to the nipple initiated by infant or mother
Proximity index: An index of the infant's relative role in maintaining proximity to the mother within a 60-cm limit. It is the percentage of approaches made by the infant to within 60-cm (divided by the total number of approaches initiated by mother and infant) minus the percentage of departures made by the infant over a 60-cm limit (divided by the total number of departures made by mother and infant) (Hinde & Atkinson, 1970)
Infant Distress-related Measures
Time tense or distressed: Percentage of point time samples infant is in an apparently tense or agitated state, as evident from muscle tonus or vocalizations (whoo calls, geckers, or screams)
Time depressed: Percentage of point time samples infant is in an apparently depressed state, as evident from typical slouched posture and social withdrawal (e.g., Kaufman & Rosenblum, 1967)
Tantrum rate: Frequency per hour of episodes in which the infant displays intense geckering and screaming accompanied by violent body movements (e.g., throwing self on ground, flailing, kicking the air)
Distress call rate: Frequency per hour of geckering and screaming
Time passive: Percentage of time samples awake but inactive or involved in autogrooming
Infant Social Interaction with Individuals Other Than the Mother
Time social: Percentage of time samples involving any nonagonistic social interaction (e.g., allogrooming, social play, inspection of body parts, exchange of affiliative gestures). Excludes mere proximity or passive contact
Time spent playing: Seconds per hour spent in vigorous social play involving contact and/or play-fighting (e.g., Symons, 1978)
Play bouts: Frequency per hour of bouts of vigorous social play
Play initiation rate: Frequency per hour at which the infant initiates vigorous social play
Play invitation rate: Frequency per hour at which another individual initiates vigorous social play with the infant
Play refusal rate: Proportion of play invitations to the infant which are not followed by play
Play partners: Total number of individuals the infant plays with divided by the number of hours observed
Time others groom: Seconds per hour in which an individual grooms the infant
Time infant grooms others: Seconds per hour in which the infant grooms another individual
Partners that groom: Total number of individuals that groom the infant divided by the number of hours observed
Partners infant grooms: Total number of individuals that the infant grooms divided by the number of hours observed

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possible effects of infant age, infant sex, and social group together. Because we found few significant differences between infants in different social groups during the mother's first estrous period (see below), we dropped social group from the design in subsequent analyses for other independent variables. Where significant associations were found between independent variables and behavior during estrus, we attempted to determine whether these associations represented differential responses by subjects to the events surrounding the mother's resumption of mating or simply preexisting differences by calculating change scores (i.e., by subtracting baseline scores from estrous period scores) and by subjecting them to ANCOVA analysis. We chose this method because, unlike the repeated-measures methods available in SPSS, it allowed us to identify differential responses to the mother's resumption of mating as a function of a continuous independent variable (covariate) (e.g., infant's age when the mother resumed mating or the number of close kin), while simultaneously controlling for one or more discrete factors. In all ANCOVA tests, factors and covariates were entered simultaneously. Hence, all "effects" were adjusted for all other independent variables in the design. All statistical tests were two-tailed. With a few exceptions, only significant ($p < .05$) results are presented.

Results

General Description of Events

The infants ranged in age from 16 to 34 weeks, $M \pm SD = 23.9 \pm 4.7$, when their mothers resumed mating. During their first estrous periods, mothers typically formed a consortship or series of consortships with one to several males for periods lasting from several minutes to several days (see Chapais, 1983). Most male consorts were infrequent associates or strangers to the infant, and infants often appeared to be frightened by them. In addition, mothers appeared to alter their relationships with their infants abruptly. In milder forms, these changes consisted of a sort of "benign neglect" by the mother as she shifted her attention from the infant to her mating partners. In more extreme cases, the mother rejected contact with the infant for hours at a time. Although some mothers remained within the center of their social group and continued to associate with close kin, many spent large amounts of time at the periphery or outside the group, presumably to avoid interference from males other than their consorts. In some cases, infants had difficulty keeping up with rapidly moving mothers and consorts, and they became physically separated from them and even from the rest of the group for periods of time ranging from a few minutes to several hours. Most infants appeared to react to these events with increased disturbance. In extreme cases, infants threw tantrums around the mother, attempted to cling to her as she mated, withdrew from social interaction with others, and emitted distress calls almost constantly.

Changes in Mother-Infant Relationships and Infant Distress

Comparisons of mother-infant interaction measures (Table 1) for the baseline and estrous periods confirmed that infants experienced dramatic changes in their relationships with their mothers when their mothers resumed mating, $F(1, 12) = 6.72$, $p = .014$, repeated-measures MANOVA. In particular, infants spent more time separated from their mothers during their mothers' estrous periods than they had previously: Mothers spent significantly more time out of their infants' sight, more time at a distance greater than 5 m, and less time within touching distance (< 60 cm) of their infants (Table 2). When mothers and infants were near one another, they spent significantly less time in all forms of bodily contact, including nursing and carrying. In addition, mothers initiated significantly smaller proportions of their nipple contacts than they had done previously. This pattern of changes suggests that changes in mothers, rather than infants, were primarily responsible for changes in mother-infant relationships (see Hinde, 1969); when mothers resumed mating, they abruptly sought less contact with their infants.

The period during which mothers resumed mating was also characterized by dramatic increases in infant contact-seeking and distress-related behavior (Table 1), $F(1, 4) = 8.56$, $p = .001$, repeated-measures MANOVA. Infants had tantrums, attempted to gain access to the nipple, and emitted distress calls at higher rates than they had previously (Table 2). Rates with which infants actually achieved nipple contact increased, and maternal rejection rates decreased, suggesting that infants may have achieved some degree of success in gaining access to the nipples of otherwise reluctant mothers.

Other indications that infants experienced more distress when their mothers re-

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TABLE 2

MEANS ± SE FOR MEASURES OF MOTHER-INFANT INTERACTION AND INFANT DISTRESS DURING THE BASELINE PERIOD AND THE MOTHER'S FIRST ESTROUS PERIOD

Measure	Baseline $M \pm SE$	First Estrus $M \pm SE$	p[a]
% time mother is out of sight	5.0 ± .0	14.2 ± .0	.000
% time mother is >5 m	19.2 ± .0	30.4 ± .0	.000
% time mother is <60 cm	11.4 ± .0	8.2 ± .0	.007
Time mother carries, minutes per hour	6.7 ± 1.4	2.7 ± 1.3	.034
% time mother is in contact	45.8 ± .0	37.3 ± .0	.032
% time on nipple	36.9 ± 2.4	29.4 ± 3.0	.037
Proportion of maternally initiated nipple contact	.21 ± .03	.09 ± .03	.002
Nipple attempts per hour	9.4 ± .8	12.0 ± 1.0	.020
Nipple contacts per hour	8.3 ± .7	10.8 ± 1.0	.011
Rejection rate	.27 ± .02	.18 ± .03	.019
Distress calls per hour	5.6 ± .9	14.7 ± 3.0	.001
Tantrums per hour	.19 ± .07	.73 ± .20	.011
% time tense or depressed[b]	5.0 ± .0	10.8 ± .0	.049
% time passive	7.0 ± .0	12.7 ± .0	.008
No. partners infant grooms	.09 ± .04	.29 ± .10	.047

[a] T tests for related samples, $N = 23$ infants.
[b] Sum of % time tense or distressed and % time depressed.

sumed mating were significant increases in the time infants were "passive" (sitting awake but inactive) and in apparently tense or depressed states (Table 2). Three infants were observed sitting in withdrawn, hunched postures similar to those of "depressed" infants in captive mother-infant separation studies (e.g., Kaufman & Rosenblum, 1967). In two of these cases, the percentages of time spent in these postures were small (3.8% and 11.7%). The third infant spent 44.0% of her time in this posture, but 90.9% of this time was after the fourth day of her mother's estrus when the infant was wounded on her arm. This infant died 2 days after being wounded. Daily plots of infant distress-related behavior revealed no evidence of regular stages of response (e.g., tense or agitated behavior followed by passive or "depressed" behavior followed by apparent recovery; see Bowlby, 1969; Kaufman & Rosenblum, 1967).

Although there were no overall trends for infants to engage in more or less social interaction with individuals other than the mother (Table 1) when mothers resumed mating, $F(1, 11) = 1.35$, N.S., repeated-measures MANOVA, some infants showed dramatic decreases in the amount of time they spent in social play. Two infants ceased playing altogether, and four played less than half the amount of time they had during the baseline period. In addition, although few infants spent substantial amounts of time grooming other individuals during the baseline period (only six groomed at all and none groomed more than 1 min per hour), some began doing so at modest rates during the estrous period (11 groomed, three of whom groomed more than 1 min per hour). This finding suggests that different infants may have exhibited qualitatively different social responses to the events surrounding their mothers' resumption of mating.

Infant Age, Infant Sex, and Social Group

The possible influences of infant age, infant sex, and social group membership on responses to the mother's resumption of mating were assessed in ANCOVA designs in which infant sex and group (group F vs. T vs. Q) were entered as factors and infant age was entered as a covariate. The sample for group T had a disproportionate number of male infants (five males vs. two females), whereas males and females were equally represented in groups F and Q. Although infant age was not strongly associated with social group or infant sex, the three oldest infants were all female.

Infant age.—During both the baseline and estrous periods, younger infants made more nipple contacts and were carried more than older infants (Table 3). Several other measures of mother-infant interaction differed significantly as a function of infant age during the mother's estrous period but not during the baseline period. For example,

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TABLE 3

AGE-RELATED DIFFERENCES IN BEHAVIOR DURING BASELINE AND THE MOTHER'S FIRST ESTROUS PERIOD: *F* RATIOS FOR AGE EFFECTS

Behavioral Measure	Baseline	Estrous Period	Change Score[a]
% time mother carries	6.36*	27.30****	1.32
% time mother is in contact	3.43	26.99****	11.10***
Nipple contact per hour	4.85*	8.99**	1.29
Nipple attempts per hour	3.29	6.34*	1.18
% time on nipple	.37	28.79****	16.29***
% time >5 m from mother	1.58	5.31*	5.24*
Proportion of maternally initiated nipple contact	.84	6.91*	.37
Time infant grooms others, seconds per hour	9.02**	15.93***	3.12
Tantrums per hour	.57	4.50*	4.31*
No. of partners that groom infant	.27	9.17**	13.23**
No of partners that infant grooms	4.44*	19.80****	13.00**
No. of play partners	.13	10.77**	8.52**

NOTE.—*F* ratios refer to infant age effects, controlling for infant sex and social group membership.
[a] Change scores = estrous period scores minus baseline period scores.
* $p < .05$.
** $p < .01$.
*** $p < .001$.
**** $p < .0005$.

during estrus, younger infants spent significantly larger percentages of time in contact and on the nipple and significantly less time at a distance of >5 m from the mother than older infants. In addition, younger infants made more attempts to get on the nipple than older infants, and their mothers initiated larger proportions of the nipple contact. When differences between estrous period and baseline period scores (change scores) were examined as a function of infant age, significant age-related results were found for percentages of time on the nipple, in contact, and at a distance of >5 m. Although most infants experienced less contact with their mothers when their mothers resumed mating, infants that were relatively young when their mothers resumed mating actually spent more time in contact and on the nipple during their mothers' estrous periods than they had previously (Fig. 1).

Younger infants may have experienced higher levels of contact and suckling during estrus both because they attempted to get on the nipple more frequently and because their mothers initiated more contact. When variations in infant age were controlled through partial correlation analysis, rates of nipple contact during estrus were found to be positively correlated with the rates with which infants attempted to make nipple contact, $r(20) = .86$, $p = .001$, and with the proportion of maternally initiated nipple contacts, $r(20) = .40$, $p = .032$. However, infants that attempted to get on the nipple fre-

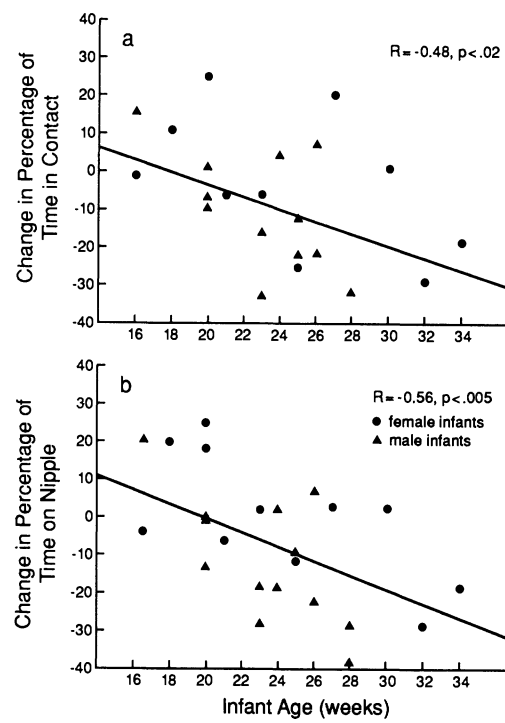


FIG. 1.—Changes in percentages of time spent (*a*) in contact with the mother and (*b*) on her nipple as a function of the infant's age when the mother resumed mating. Measures of change were calculated by subtracting baseline period scores from estrous period scores.

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quently were also punished frequently, $r(20) = .40$, $p < .032$. Hence, relatively demanding infants, regardless of age, were likely to have high rates of both nipple contact and punishment during estrus.

During baseline, there were no age-related differences in measures of infant distress behavior or social play. However, older infants spent more time grooming individuals other than the mother than did younger infants (Table 3). This difference persisted when mothers entered estrus. In addition, during their mothers' estrus, older infants had fewer tantrums than younger infants, and they played with, groomed, and were groomed by more partners. When change scores were examined, significant age-related results were found for rate of tantrums, number of grooming partners, and number of play partners. Hence, younger infants were more likely to respond to their mothers' resumption of mating by having frequent tantrums, whereas older infants apparently responded by grooming and playing with a wider selection of partners.

Sex of the infant.—During the mother's estrus, there were no sex differences in distress-related behaviors. However, there were marked sex differences in play and grooming behavior that had not been present previously (Table 4). Male infants initiated play more frequently than female infants, received more play invitations, spent more time playing, had more bouts of play, and played with a larger number of play partners. Males tended to increase the time they spent time playing from baseline levels, whereas females tended to play much less than they had previously (Table 4). Females, in contrast, spent more time grooming individuals other than the mother than did males. Male infants groomed others about as long as they had during the baseline period, whereas female infants groomed others about three times as long (Table 4). When change scores were examined, significant sex differences were found for time spent playing and for rates of play bouts and play invitations to others, suggesting that infants of each sex indeed responded differently to the events surrounding their mother's first estrus.

During estrus, but not baseline, male infants spent significantly larger percentages of time >5 m from their mothers than did female infants (Table 4). Change scores for this measure also differed significantly by infant sex, suggesting that mother-infant pairs with male infants responded differently to the mother's resumption of estrus than those with female infants, perhaps because males spent more time playing.

Social group.—During estrus, there were few differences between groups in measures of mother-infant interaction, infant distress, play, or social grooming. Only one measure, rejections, differed during estrus but not during baseline: Mothers in group T rejected their infants at higher rates than mothers in other groups, $M = .13, .15, .28$ for groups F, Q, and T, respectively, $F(2, 22) = 5.50$, $p = .015$. Change scores for rejections also showed significant group differences, $M = -.11, -.15, +.03$ for groups F, Q, and T, respectively, $F(2, 22) = 4.50$, $p = .028$. Two measures differed during baseline but not estrus: time spent playing, $M = 46.6$, 78.9, 183.6 sec per hour for groups F, Q, and T, respectively, $F(2, 22) = 10.48$, $p = .001$, and numbers of play partners, $M = 1.3$, 2.0, 3.4 for groups F, Q, and T, respectively, $F(2, 22) = 8.83$, $p = .003$. Change scores were also significantly different (time spent playing: $M = +26.9, -17.7, -71.4$ sec per hour for groups F, Q, and T, respectively, $F(2, 22) = 5.71$, $p = .013$; number of play partners: $M = +0.4, 0.0, -1.0$ for groups F, Q, and T, respectively, $F(2, 22) = 4.74$, $p = .024$).

Dominance Status

The possible differential influence of mother's dominance status on responses to the mother's resumption of mating was assessed in ANCOVA tests in which status (high vs. low) and infant sex were entered as factors and infant age was entered as a covariate. Sex and age were included because of their associations with a large number of variables (see above). In addition, female infants in this sample tended to be over-represented in the low-ranking lineages, Fisher's exact probability = .09.

There were no status-related differences in mother-infant interaction or in infant distress behavior during the baseline or estrous periods. During baseline but not estrus, infants with high-ranking mothers refused a smaller proportion of play invitations than infants with low-ranking mothers, $M_{high} = 0.21$, $M_{low} = 0.41$, $F(1, 22) = 6.23$, $p = .023$. During estrus but not baseline, infants with low-ranking mothers spent less time grooming individuals other than the mother, $M_{high} = 17.8$ sec per hour, $M_{low} = 7.8$ sec per hour, $F(1, 22) = 11.23$, $p = .004$, and were groomed by fewer partners, $M_{high} = 1.1$, $M_{low} = 0.6$, $F(1, 22) = 6.92$, $p = .017$,

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TABLE 4

DIFFERENCES BETWEEN MALE AND FEMALE INFANTS IN BEHAVIORAL MEASURES DURING BASELINE AND THE MOTHER'S ESTROUS PERIOD: MEANS ± SE AND *F* RATIOS FOR SEX EFFECT

Behavioral Measure	*M* ± SE Males	*M* ± SE Females	*F* Ratio
% time >5 m from mother:			
Baseline	19.9 ± 3.3	18.2 ± 3.4	.05
Estrus	37.6 ± 4.3	21.1 ± 6.9	5.42*
Change score	+17.7 ± 2.5	+2.9 ± 4.1	11.82**
Play initiations with others per hour:			
Baseline	3.3 ± .5	1.8 ± .4	4.11
Estrus	5.0 ± .6	1.6 ± .5	15.49***
Change score	+1.7 ± .7	−.2 ± .5	5.30*
Play invitations from others per hour:			
Baseline	3.7 ± .5	3.2 ± .6	.51
Estrus	4.2 ± .6	2.2 ± .6	5.55*
Change score	+.5 ± .7	−1.0 ± .5	2.80
Time spent playing, seconds per hour:			
Baseline	105.2 ± 19.2	92.1 ± 29.6	.22
Estrus	116.0 ± 17.7	35.8 ± 10.4	12.34**
Change score	+10.8 ± 21.8	−56.3 ± 30.0	8.87**
Play bouts per hour:			
Baseline	6.3 ± .8	4.7 ± 1.2	.44
Estrus	7.4 ± 1.0	2.5 ± .6	18.59***
Change score	+1.1 ± 1.0	−2.2 ± 1.1	9.94**
No. of play partners:			
Baseline	2.5 ± .3	1.8 ± .4	.94
Estrus	2.6 ± .3	1.3 ± .4	10.25**
Change score	+.1 ± .4	− .4 ± .5	3.35
Time infant grooms others, seconds per hour:			
Baseline	3.2 ± 2.9	8.2 ± 5.0	.18
Estrus	2.3 ± 1.4	25.0 ± 11.0	15.58***
Change score	−.9 ± 3.3	+16.8 ± 8.9	11.06**

NOTE.—*F* ratios refer to infant sex effects, controlling for social group membership and infant age. Change scores = estrous period scores minus baseline period scores.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

than infants with high-ranking mothers. In addition, during estrus, there was a significant interaction between status and sex for time spent in social play, $F(1, 22) = 5.23$, $p = .035$: male infants with high-ranking mothers played less than males with low-ranking mothers during estrus, $M_{high} = 82.2$ sec per hour, $M_{low} = 170.1$ sec per hour, but this was not the case among female infants, $M_{high} = 54.7$ sec per hour, $M_{low} = 31.0$ sec per hour. Change scores only for proportions of play refused, $M_{high} = +.26$, $M_{low} = -.16$, $F(1, 22) = 20.71$, $p = .0005$, and time spent grooming with others, $M_{high} = +11.1$ sec per hour, $M_{low} = +3.5$ sec per hour, $F(1, 22) = 11.06$, $p = .004$, were significantly different for infants of different status.

Immature Sisters and Numbers of Close Kin

Close kin were defined as group members related through the mother by at least as much as mother's half-siblings (i.e., maternal aunts, uncles, nephews, and nieces). Infants had between 3 and 34 close kin, $M \pm SD = 9.7 \pm 7.0$. Because infants that had relatively large numbers of close kin tended to have relatively more immature sisters (1–3 years of age), $r(22) = .54$, $p = .01$, these variables were analyzed together in an ANCOVA design in which the presence of immature sisters (no sisters vs. one or more) and infant sex were entered as factors and infant age and number of close kin were entered as covariates.

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There were no significant associations between the presence of immature sisters and behavioral measures during the baseline or estrous periods. During estrus, infants with relatively more close kin were groomed by more partners than infants with fewer close kin, $F(1, 22) = 5.34$, $p = .034$, a tendency that was already present (but nonsignificant) during baseline, $F(1, 22) = 4.16$, $p = .057$.

Mother's Responsiveness to Infant Distress

The findings that younger infants had more tantrums than older infants during estrus (Table 3) and actually achieved more contact with the mother when she was in estrus than previously (Fig. 1) suggest that mothers may have responded to tantrums by allowing infants more contact. Indeed, when infant age was controlled through partial correlation analysis, increases in rates of tantrums were positively correlated with both contact time during estrus, $r(20) = .32$, $p = .05$, and with increases in contact time, $r(20) = .42$, $p = .027$. There was no evidence that high rates of distress calling or other distress-related behavior were related to increased probabilities of contact with the mother when she resumed mating or to high levels of contact either during the baseline or estrous periods. On the contrary, the rates of distress calling during estrus were negatively correlated with time on the nipple, $r(20) = -.44$, $p = .02$, and positively correlated with maternal rejection rates, $r(20) = .52$, $p = .006$, and with the percentage of time mothers spent out of sight of infants, $r(20) = .39$, $p = .038$. This suggests either that mothers were more likely to reject calling infants or that distress calls were responses to maternal rejection and/or separation.

Although mothers often ignored their infants' tantrums, four of eight mothers of the youngest infants (≤21 weeks) responded consistently to tantrums by initiating or allowing their infants to establish nipple contact within 2 min of a tantrum (consistently responsive or CR mothers). Six other mothers did so inconsistently ($n = 4$) or not at all ($n = 2$). The infants of the 13 remaining mothers displayed too few tantrums (<2) for analysis. During estrus, CR mothers spent more time in contact with their infants than other mothers of comparably aged infants, $Z = -2.02$, $p = .043$, Mann-Whitney U test, carried them more, $Z = -2.31$, $p = .021$, Mann-Whitney U test, and initiated larger proportions of nipple contact, $Z = -2.02$, $p = .043$, Mann-Whitney U test. Indeed, the four CR mothers had the first, second, third, and fifth highest scores for proportions of maternally initiated nipple contact during estrus of all 23 mothers. Interestingly, during their mothers' estrus, infants of CR mothers had more tantrums than comparably aged infants of less responsive mothers, $M = 2.0$ per hour and 0.5 per hour for CR and other infants, respectively, $Z = -2.03$, $p = .043$, Mann-Whitney U test. This finding suggests either that mothers responded consistently to young infants giving high levels of tantrums or that some mothers encouraged young infants to have frequent tantrums during the estrus period by responding consistently to their tantrums. The latter explanation is more likely, because mothers' response styles tended to be similar during the estrous and baseline periods, $p = .062$, $n = 5$, sign test, when tantrum rates were low.

Discussion

The results of this study support the hypothesis that basic parallels exist between the events surrounding a rhesus monkey mother's resumption of mating in the field and the separation of rhesus infants from mothers in the laboratory. The first and most basic parallel is that infants in both situations experience physical separation from the mothers. During their first intensive mating periods, Cayo Santiago mothers spent nearly three times the amount of time out of sight of their infants than in the weeks immediately before. When in sight of their infants, they spent more time at a distance from them than previously, and in most cases, they spent less time in every type of contact. In addition, mothers initiated much smaller proportions of the nipple contact with their infants. In this sense, infants experienced increased periods of both physical and psychological separation from their mothers as a result of their mothers' decreased propensities to seek contact and proximity with them.

Second, infants in both situations displayed clear increases in distress-related behavior. When Cayo Santiago mothers resumed mating, several measures of distress-related behavior in their infants increased. The finding that rates of distress calling during estrus were correlated with rates of maternal rejection and with the amount of time mothers were out of sight of their infants suggests that infant distress during the mother's estrous period was

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indeed related to disruptive changes in mother-infant relationships.

As in laboratory studies of mother-infant separation in rhesus, most infants' reactions were characterized by increased agitation and distress vocalization. A minority (n = 3 or 13%) showed periods of withdrawn and apparently depressed behavior during their mothers' estrus. Although no predictable stages of distress responding were observed, such stages are not always seen in captive studies (Lewis, McKinney, Young, & Kraemer, 1976). The facts that (*a*) few infants showed signs of depression, (*b*) those who did, did so for only small amounts of time (except for one seriously injured infant), and (*c*) infants showed no overall decreases in the amount of social activity they engaged in suggest that infant distress reactions were generally mild. Given that distress reactions in separated captive infants are related to the novelty and length of separation (Chappell & Meier, 1975; Spencer-Booth & Hinde, 1971) and to the degree to which mother-infant relationships are disrupted (Hinde & McGinnis, 1977; Suomi, Collins, Harlow, & Ruppenthal, 1976), these generally mild responses may have arisen from the incomplete nature of separation in the field; infants almost always established contact with their mothers at least daily. Another contributing factor may have been the brevity of mothers' separations from the female core of the group. In captive studies, infant distress reactions following reunion with the mother are generally more severe when the mother is removed from the social group and the infant remains than vice versa (Hinde & McGinnis, 1977). Unlike the captive mothers, those in the present study had little need to reestablish relationships with other females when they returned to their infants and to the female core. As a result, they may have been in a better position to meet the increased demands of their infants. Indeed, during these "reunions," mothers often appeared fatigued and nursed their infants passively during long naps. Whether the occurrence and duration of these reunion periods contributed to decreases in maternal rejection rates from baseline levels is not yet known. Finally, infants were apparently able to influence their mothers' behavior to some extent. When their mothers resumed mating, both infants' attempts to gain access to the nipple and actual rates of nipple contact increased from baseline levels. Moreover, high levels of attempts during estrus were associated with high levels of nipple contact. Some infants, particularly young ones, were also apparently able to elicit extra carrying and contact by throwing tantrums, although no other measures of distress were related to increased probabilities of contact. As in captive studies of infants within a similar age range (e.g., Hinde & McGinnis, 1977; Mineka & Suomi, 1978), other measures of distress were unrelated to infant age.

The infants' generally mild distress responses were unlikely to be due to the high availability of close associates other than the mother. The absence of differences in distress-related behavior between infants with and without older immature sisters or between infants with many and few close kin argues against this explanation. In harmony with this interpretation, Hinde (1977) found no relation between level of distress during separation and amounts of interaction with other group members in socially living, captive rhesus infants (see also Reite, Seiler, & Short, 1978), although captive infants of some primate species have been found to respond with less distress to separation from the mother when they were able to interact with other close associates (Mineka & Suomi, 1978; Robertson & Robertson, 1971; Rosenblum & Kaufman, 1968; Schlottmann & Seay, 1972).

It could be argued that the changes in mother-infant relationships observed here were simply reflections of long-term trends toward mutual independence between mothers and infants. However, the findings that older infants experienced larger decreases in contact with their mothers than younger infants (and that younger infants actually achieved more contact with their mothers) when mothers resumed mating argue against this interpretation; during a given time period, mother-infant contact in rhesus monkeys generally decreases more rapidly in younger infants than in older infants (Berman, 1980a; Hinde & Spencer-Booth, 1967).

The results of the analyses for sources of individual variation in responses to the mother's resumption of mating revealed a number of discontinuities between the baseline and estrous periods. For example, some differences related to group membership and to maternal dominance status were evident during baseline periods but were not sustained when mothers resumed mating. Conversely, several differences related to the infant's sex were observed only during

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the mother's estrous periods. These discontinuities suggest that, in addition to general decreases in mothers' propensities to have contact with their infants when they resumed mating, there was a radical restructuring of infant social relationships. The extent to which new patterns of interaction were sustained or augmented during the remainder of the mating season is not yet known. Hence, at this point, the potential importance of these discontinuities for the infant's future development of social relationships is unclear.

One of the most striking discontinuities concerned the appearance of strong sex differences in infant behavior when the mother resumed mating. Female infants apparently reacted to their mothers' resumption of mating by playing less than they had previously and by engaging in more social grooming with individuals other than the mother. Males, particularly sons of low-ranking mothers, continued to play at high rates. Differences in amounts of vigorous social play between males and females have been previously reported to develop in late infancy in free-ranging rhesus monkeys (Berman, 1980b) and to continue at least until adolescence and early adulthood (Breuggeman, 1978; Levy, 1980; Symons, 1978). Sex differences in grooming have been reported to appear in free-ranging rhesus yearlings and to continue throughout life (Missakian, 1974), as females become progressively more integrated into the adult female core of the group and as males become gradually more integrated into play relationships with peers and more peripheral from the female core. Although such sex differences in play and grooming have been widely reported in many species of nonhuman primates (e.g., Walters, 1987), their ontogeny has not been previously associated with the mother's resumption of mating. Hence, the results of this study raise the possibility that the events accompanying the mother's resumption of mating may in some way facilitate the development or manifestation of species-normative sex differences. It is tempting to speculate that the larger increases in time spent playing by sons of low-ranking mothers relative to sons of high-ranking mothers on Cayo Santiago may foreshadow their typically earlier peripheralization from the female core of the group (Colvin, 1983; Sade, 1971).

In summary, the results of this study reveal several basic parallels between the behavioral responses of rhesus infants to their mothers' resumption of mating in the field and to forcible separation from mothers in captivity. In both situations, infants experience increased separation from the mother and respond with dramatic increases in distress behavior. To this extent, studies of mother-infant separation in captivity may be said to model species-normative developmental processes in the field. To the extent that differences may exist between responses (e.g., in the severity of distress responses and in the amount of social interaction of infants with group members), hypotheses can be made about the factors that influence responses in the field and about the possible coping mechanisms that infants may be able to utilize in complexly organized social groups. Finally, further studies of the mother's resumption of mating in the field may shed light on the ontogeny of species-normative sex differences in social behavior and perhaps on other long-term developmental consequences of early separation experiences.

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